

ISYE 4031: Homework 1 Report

Jordyn Velde

PART 1: Descriptive Summaries and Uncertainty

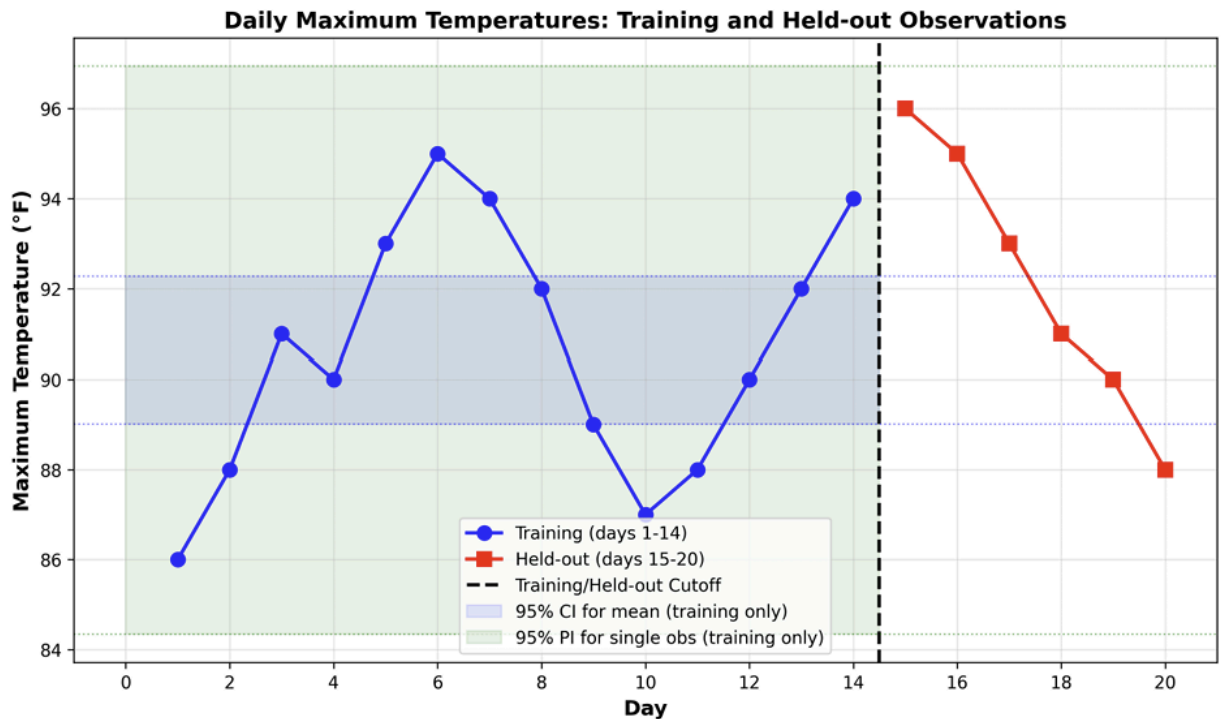
Unit of Analysis: Individual day | Response: Daily max temperature (F)

Information Cutoff: End of day 14 (14 training observations)

Forecast Target: Daily max temperatures for days 15-20

1a. Descriptive Plot

Plot shows 20 daily temperatures in order. Blue circles (days 1-14) are training data. Red squares (days 15-20) are held-out test data. Vertical line marks training/test cutoff.



1b. Descriptive Statistics (Training Data Only)

Mean: 90.6429 F | Median: 90.5000 F | Range: 9.0000 F

Sample Variance: 7.9396 F² (n-1=13) | Std Dev: 2.8177 F | IQR: 4.5000 F

1c. Confidence Interval for Population Mean

Formula: $\bar{x} \pm t(\alpha/2, n-1) * (s / \sqrt{n})$

$\bar{x} = 90.6429$, $s = 2.8177$, $n = 14$, $SE = 0.7531$, $t(0.025, 13) = 2.1604$

Margin of error = $2.1604 * 0.7531 = 1.6269$

Result: 95% CI = [89.0160, 92.2698] F

Interpretation: ~95% of repeatedly constructed CIs would contain the true population mean.

1d. Prediction Interval for Single Observation

Formula: $\bar{x} \pm t(\alpha/2, n-1) * s * \sqrt{1 + 1/n}$

$\sqrt{1 + 1/14} = 1.0351$, Margin of error = $2.1604 * 2.8177 * 1.0351 = 6.3010$

Result: 95% PI = [84.3419, 96.9438] F

PI is wider than CI because it accounts for both mean uncertainty AND natural observation variation.

PART 2: Transparent Forecast Baselines

2a & 2b. Baselines and Evaluation

Mean Baseline: Use training mean (90.6429 F) for all forecasts

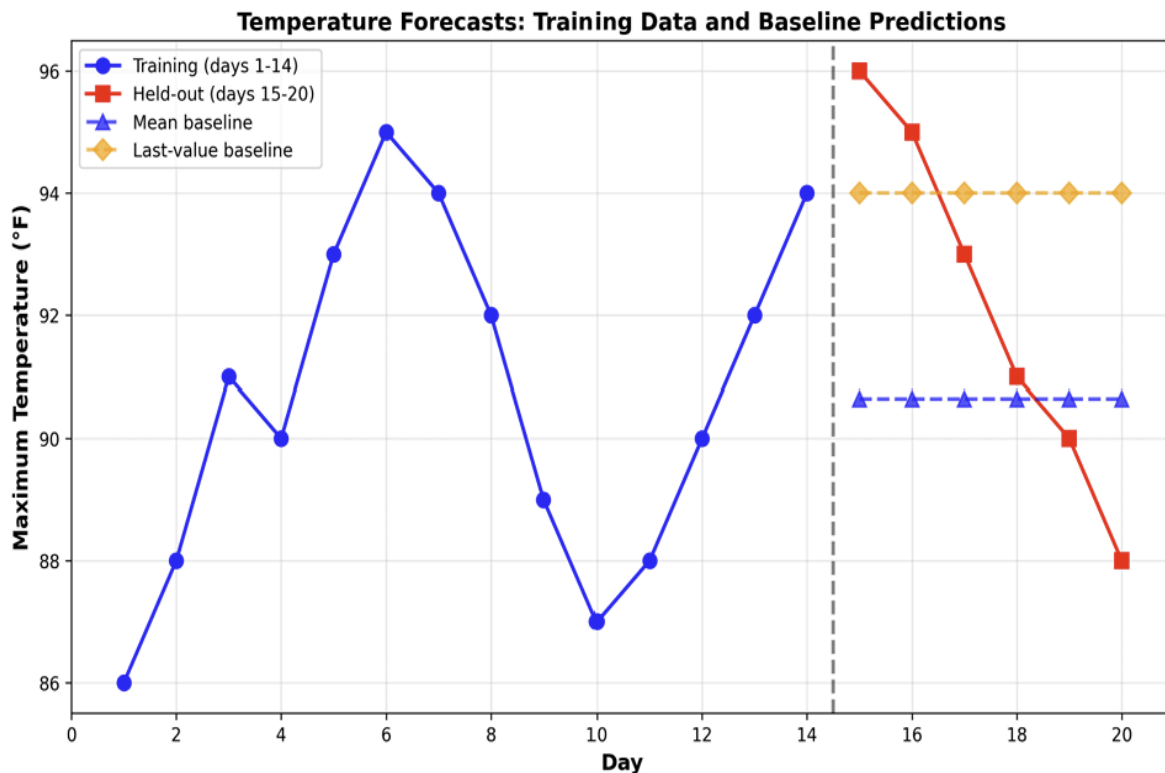
Last-Value Baseline: Use final training value (94.00 F) for all forecasts

Evaluation Results:

Mean Baseline: MAE = 2.6190 F, RMSE = 3.1824 F

Last-Value: MAE = 2.8333 F, RMSE = 3.3417 F

Winner: Mean baseline performs better on both metrics.



2c. Independence of Errors

Are errors independent? NO. Reasons: (1) Temperature autocorrelation - warm days follow warm days. (2) Seasonal patterns create systematic dependence. (3) Time series data violates i.i.d. assumption.

2d. Recommended Baseline

Choice: MEAN BASELINE. Justification: Simple, better performance (RMSE 3.18 vs 3.34), reproducible, and theoretically optimal under squared-error loss.

PART 3: One-Sample Inference

3a. Hypothesis Test

H0: $\mu = 155$ F | Ha: $\mu \neq 155$ F | $\alpha = 0.01$

Data: $n=10$, $\bar{x}=154.2$ F, $\sigma=1.5$ F (known)

Test Statistic: $z = (154.2 - 155) / (1.5 / \sqrt{10}) = -1.6865$

Critical Value: $z(0.005) = 2.5758$

Decision: $|z|=1.6865 < 2.5758$, so FAIL TO REJECT H0

Conclusion: Insufficient evidence that mean differs from 155 F at $\alpha=0.01$.

3b. Two-Sided P-value

p-value = $2 * P(Z > 1.6865) = 0.0917$

Interpretation: Probability of observing sample mean at least as extreme as 154.2 F when H0 is true.

Since $0.0917 > 0.01$, we fail to reject H0.

3c. Type II Error (Beta)

If true mean is 150 F: Acceptance region = $[153.78, 156.22]$ F

Beta = $P(\text{fail to reject H0} \mid \text{true mean} = 150) \sim 0.0000$

High power: Effect size (5 F difference) is very large relative to standard error.

3d. Python Verification

Calculations verified with `scipy.stats`:

`z_stat = -1.686548` [verified] | `z_critical = 2.575829` [verified]

`p_value = 0.091690` [verified] | `beta = 0.000000` [verified]

PART 4: Agent Audit and Automated Tests

4a.

The proposed code contains five distinct problems. First, it computes statistics using the full 20-observation series instead of only the first 14 training observations. This allows held-out values to leak into the analysis and violates the information cutoff. Second, it uses $\text{ddof}=0$ instead of $\text{ddof}=1$ for the sample variance and standard deviation. This divides by n instead of $n-1$ and underestimates the training-data variability. Third, it computes the prediction interval as if it were the confidence interval for the mean. The correct prediction interval is $\bar{x} \pm t(\alpha/2, n-1) * s * \sqrt{1 + 1/n}$. This interval is wider because it includes both uncertainty in the estimated mean and natural variation in a new observation. Fourth, it uses an inappropriate normal critical value instead of the t distribution for the small sample size. With $n=14$, the t distribution with $n-1=13$ degrees of freedom should be used. Fifth, the last-value baseline uses the final held-out observation through $x[-1]$ instead of the final training observation through $\text{train}[-1]$. This uses future information and makes the forecast invalid. The proposed code also lacks clear reproducibility documentation, so the corrected implementation documents that the calculations are deterministic and use no random procedures.

4b.

The corrected implementation computes all descriptive statistics, confidence intervals, prediction intervals, and forecasts using only the first 14 observations in the training set. It uses $\text{ddof}=1$, so the sample variance and standard deviation use the correct $n-1$ denominator. The confidence interval uses the formula $\bar{x} \pm t(\alpha/2, n-1) * s / \sqrt{n}$, and the prediction interval uses the formula $\bar{x} \pm t(\alpha/2, n-1) * s * \sqrt{1 + 1/n}$.

The two forecast baselines remain separate. The mean baseline repeats the training mean for all six future observations, while the last-value baseline repeats the final training value, 94 degrees Fahrenheit, for all six future observations. The held-out observations are used only to evaluate the forecasts with MAE and RMSE, not to construct them.

4c.

The automated tests verify that changing the held-out observations does not change the training statistics, confidence interval, prediction interval, or baseline forecasts. They verify that the sample standard deviation uses the $n-1$ denominator, that the prediction interval is wider than the confidence interval, and that the theoretical confidence-interval/prediction-interval width relationship is correct. The tests also verify reproducibility, correct separation of the training and held-out data, correct forecast lengths, correct hypothesis-test calculations, and the Type II error calculation. The complete test suite contains 12 tests, and all 12 tests pass.

Summary and Key Guarantees

PART 1: Using the 14 training temperatures, the 95% confidence interval was [89.02, 92.27] degrees Fahrenheit, and the 95% prediction interval was [84.34, 96.94] degrees Fahrenheit. The prediction interval is wider because it includes both mean uncertainty and individual temperature variation.

PART 2: The mean baseline had a lower RMSE than the last-value baseline, 3.18 versus 3.34 degrees Fahrenheit, so the mean baseline was recommended.

PART 3: The hypothesis test produced $z = -1.69$ and $p = 0.092$. Since $p > 0.01$, we failed to reject $H_0: \mu = 155$ degrees Fahrenheit. The Type II error probability when the true mean is 150 degrees Fahrenheit was approximately zero.

PART 4: The audit identified five problems: data leakage, incorrect ddof, an incorrect prediction interval formula, use of a normal instead of t critical value, and use of a held-out value for the last-value forecast. The corrected implementation uses training data only, separates both baselines, and is included in `analysis.py` and `test_analysis.py`. All 12 automated tests passed.

KEY GUARANTEES:

- No held-out data are used to construct statistics or forecasts.
- Sample variance uses the $n - 1$ denominator.
- Confidence and prediction intervals use the correct formulas.
- Results are deterministic and reproducible.
- All 12 automated tests passed.